Filtering in EvalData
A Small Algorithm Note

EvalData Repository Example

April 2026

1 Purpose

This document is a short formal note for the filtering methods currently visible in the EvalData analysis workspace.

The examples are intentionally reproducible. They use the built-in spectral reference demo dataset exposed by the app's demo menu.

Use the practical workflow pages first. Use this note when you want the algorithm-level description of what the filtering controls actually do.

2 Method Groups

EvalData currently exposes two filtering groups in the analysis workspace:

- **Simple Filtering**: keep values inside a selected range and write the result to a column.
- **Signal Processing**: create a smoothed or trend-removed version of a numeric signal.

Both groups operate on the workspace's working dataframe copy. They do not edit the original loaded dataset in place.

3 Simple Filtering

Simple filtering is a masking operation. For a source signal $x[n]$ and a range $[a, b]$, the output can be written informally as

$$y[n] = \begin{cases} x[n], & a \le x[n] \le b, \\ \text{NaN}, & \text{otherwise.} \end{cases}$$

In EvalData, the user may supply only a minimum or only a maximum. Missing values can optionally be preserved.

This is useful when you want to hide out-of-range values without dropping rows from the rest of the dataframe.

4 Smoothing Filters

4.1 Moving Average

The moving average is a local low-pass smoother. For an odd window width $W = 2m + 1$,

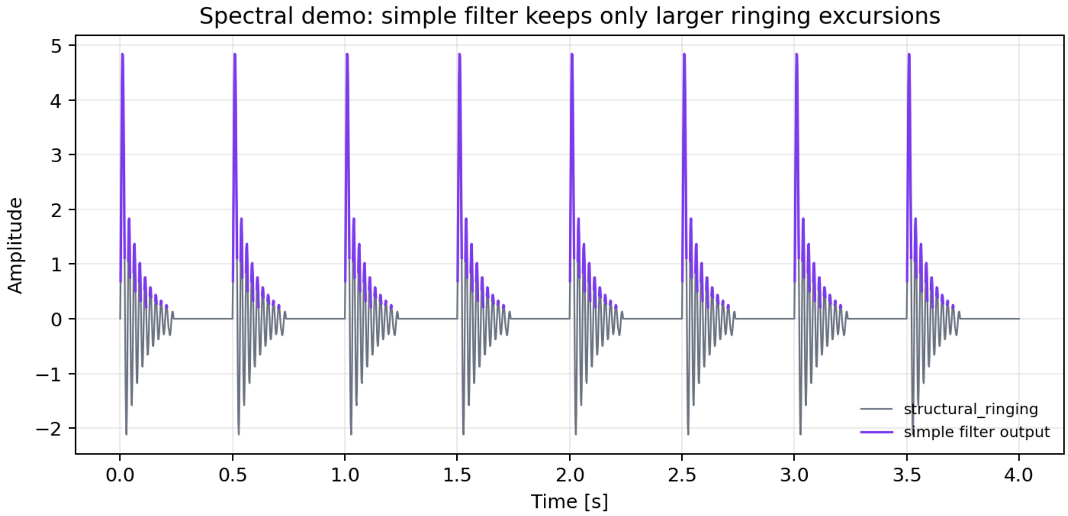

Figure 1: Simple filtering on the built-in spectral demo. The example keeps only larger values of the known `structural_ringing` component. The row count stays unchanged; only the output column is masked.

$$y[n] = \frac{1}{W} \sum_{k=-m}^{m} x[n+k].$$

In EvalData this is implemented as a centered rolling mean.

Use it when the signal is noisy but the underlying variation is slower than the sample-to-sample fluctuations.

4.2 Median Filter

The median filter replaces each point by the median of its local neighborhood:

$$y[n] = \mathrm{median}\{x[n-m], \ldots, x[n+m]\}.$$

This is less sensitive to isolated spikes than a moving average.

Use it when you want smoothing that is more resistant to outliers.

4.3 Exponential Smoothing

Exponential smoothing is recursive:

$$y[n] = \alpha x[n] + (1-\alpha) y[n-1], \qquad 0 < \alpha \leq 1.$$

Smaller α produces heavier smoothing and more lag.

5 High-Pass Filtering

The current high-pass option in EvalData is implemented as

$$y[n] = x[n] - \mathrm{LP}_W\{x[n]\},$$

where LP_W is the moving-average trend computed with the chosen window size.

This is a pragmatic trend-removal method rather than a full designed digital high-pass filter with a specified frequency response.

Use it when you want to suppress slow drift and emphasize faster variations.

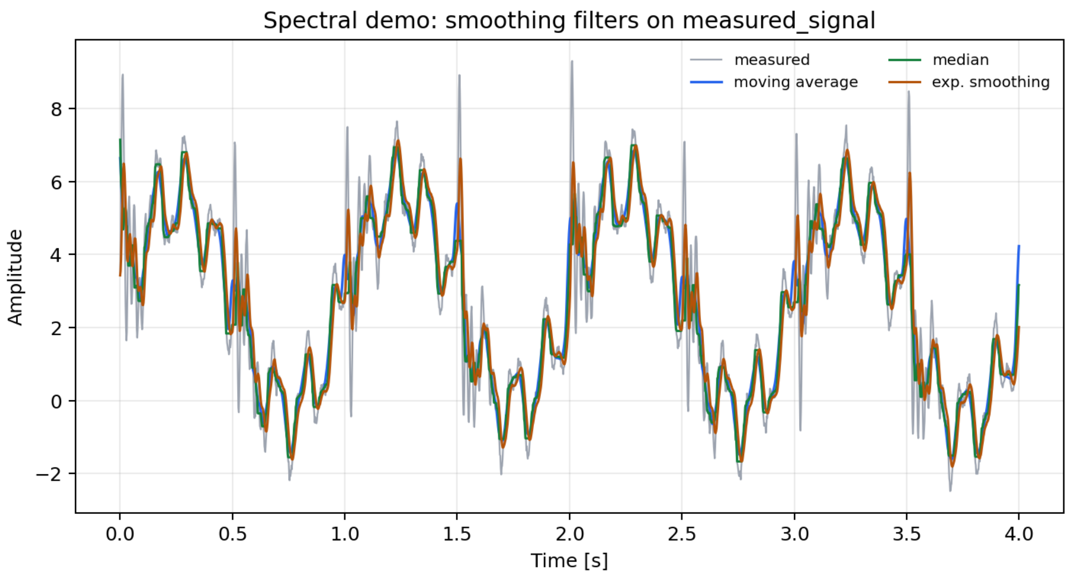

Figure 2: Three smoothing filters applied to the deterministic `measured_signal` channel from the built-in spectral demo. The comparison is reproducible because the demo signal is fixed.

6 What the Demo Shows

The spectral reference demo contains several ingredients on purpose:

- slow drift near 0.15 Hz,
- clean periodic content near 1.0 Hz, 7.5 Hz, and 18.0 Hz,
- repeated impacts,
- decaying structural ringing near 42 Hz,
- seeded broadband noise.

That makes it a useful control case for filtering documentation:

- smoothing can reduce the seeded noise,
- high-pass filtering can suppress the slow drift,
- simple filtering can isolate larger ringing excursions.

7 Practical Interpretation Limits

These methods are useful, but they have limits:

- stronger smoothing also blurs short events,
- median and moving-average filters do not preserve the same features equally well,
- exponential smoothing introduces lag,
- the current high-pass option should not be described as a precise frequency-domain filter design,

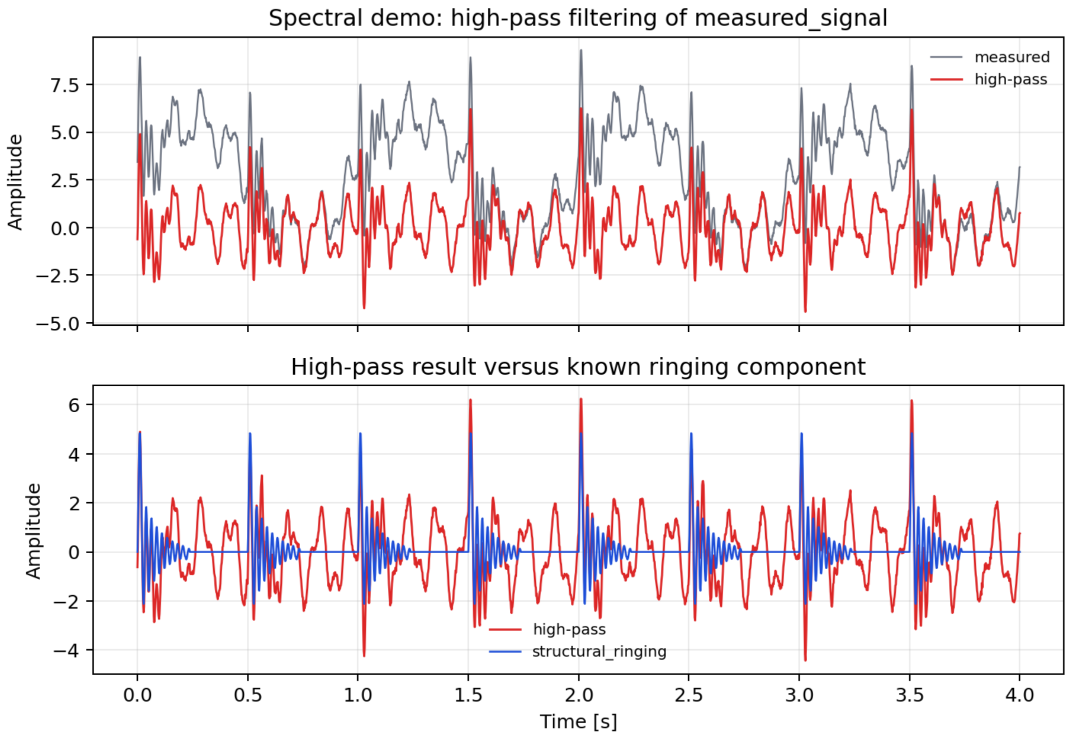

Figure 3: High-pass filtering on the built-in spectral demo. The lower panel compares the high-pass result against the known `structural_ringing` component included in the synthetic dataset.

- a masked simple-filter output is not the same thing as row-wise trimming.

Those distinctions matter because the user-visible controls are operationally simple, but their outputs answer different questions.

8 Conclusion

EvalData's current filtering tools are lightweight and practical. Simple filtering is a masking tool. The signal-processing filters are smoothing or trend-removal tools. The built-in spectral demo gives a reproducible reference case for showing the difference between those behaviors without relying on private measurement data.